

Dust growth assisted by streaming instability

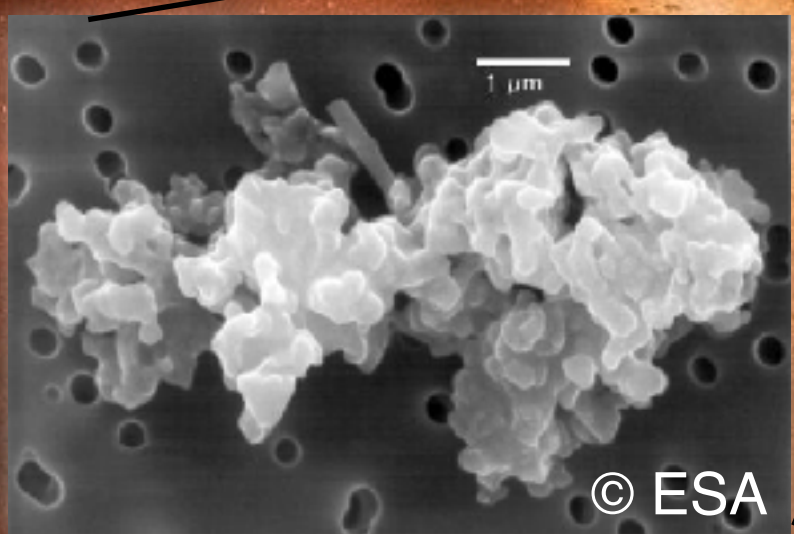
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Collaborators: Hidekazu Tanaka, Kota Wada (Tohoku University)

Tominaga & Tanaka (2023, 2025), Wada et al. in prep.

How do planetesimals form?

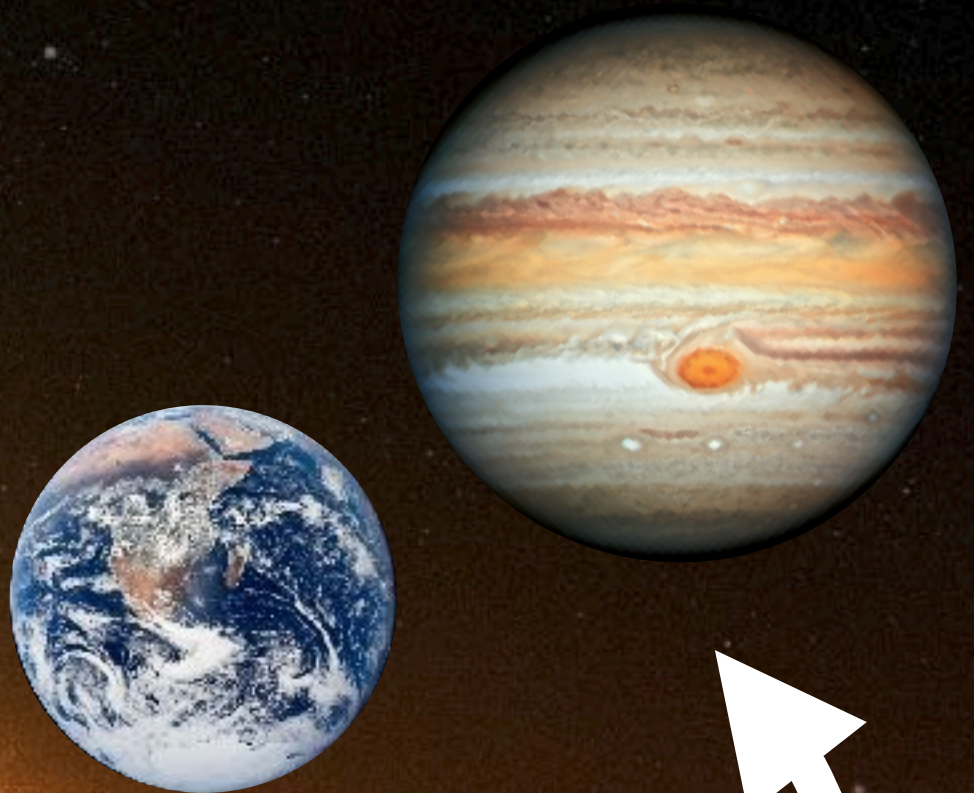
Dust grains



Collisional growth

Hydrodynamic instabilities +
Gravitational collapse

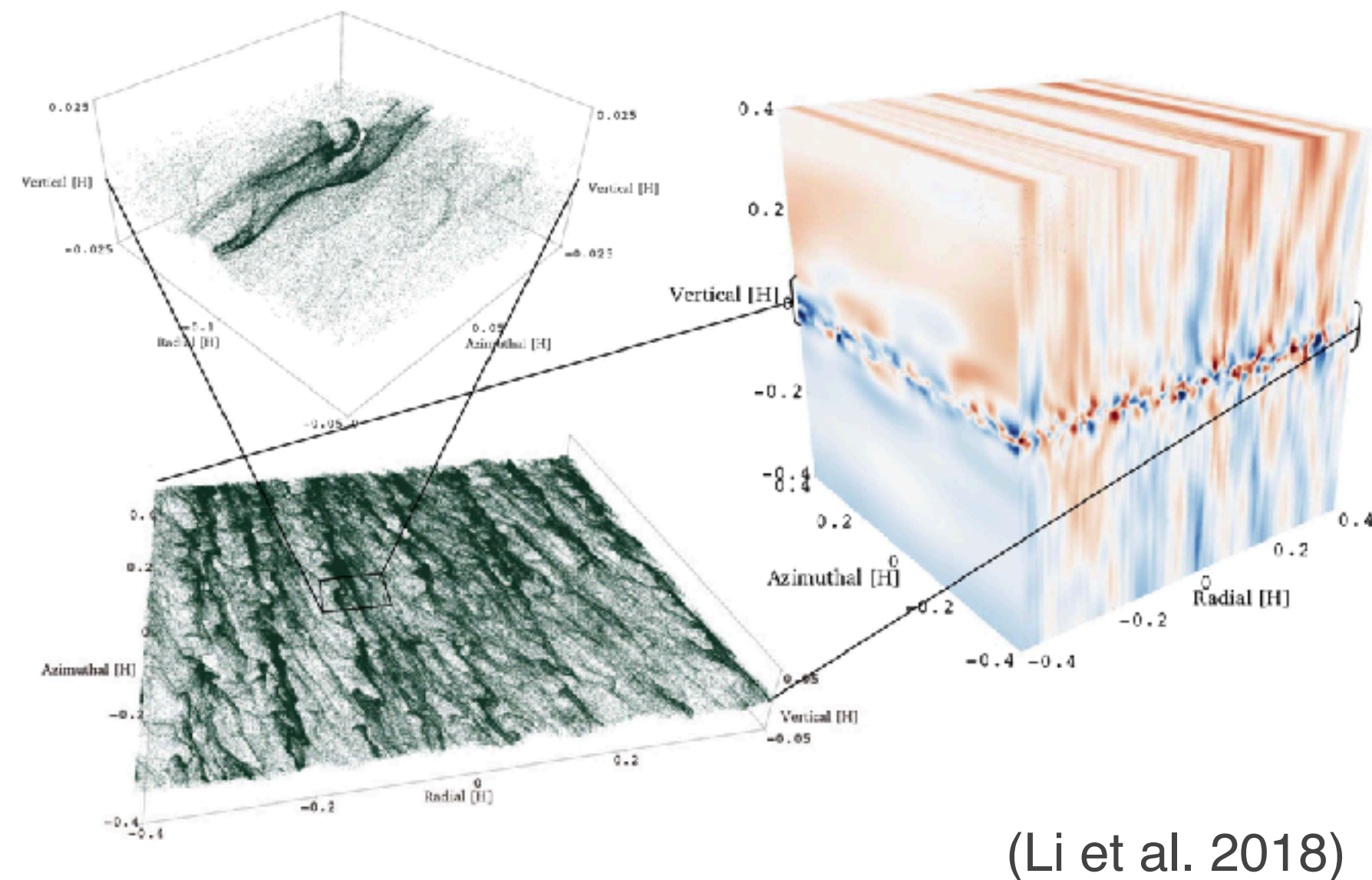
Planetesimals



Bouncing/Fragmentation/Radial drift

Some dust-gas instabilities promoting planetesimal formation

Streaming instability



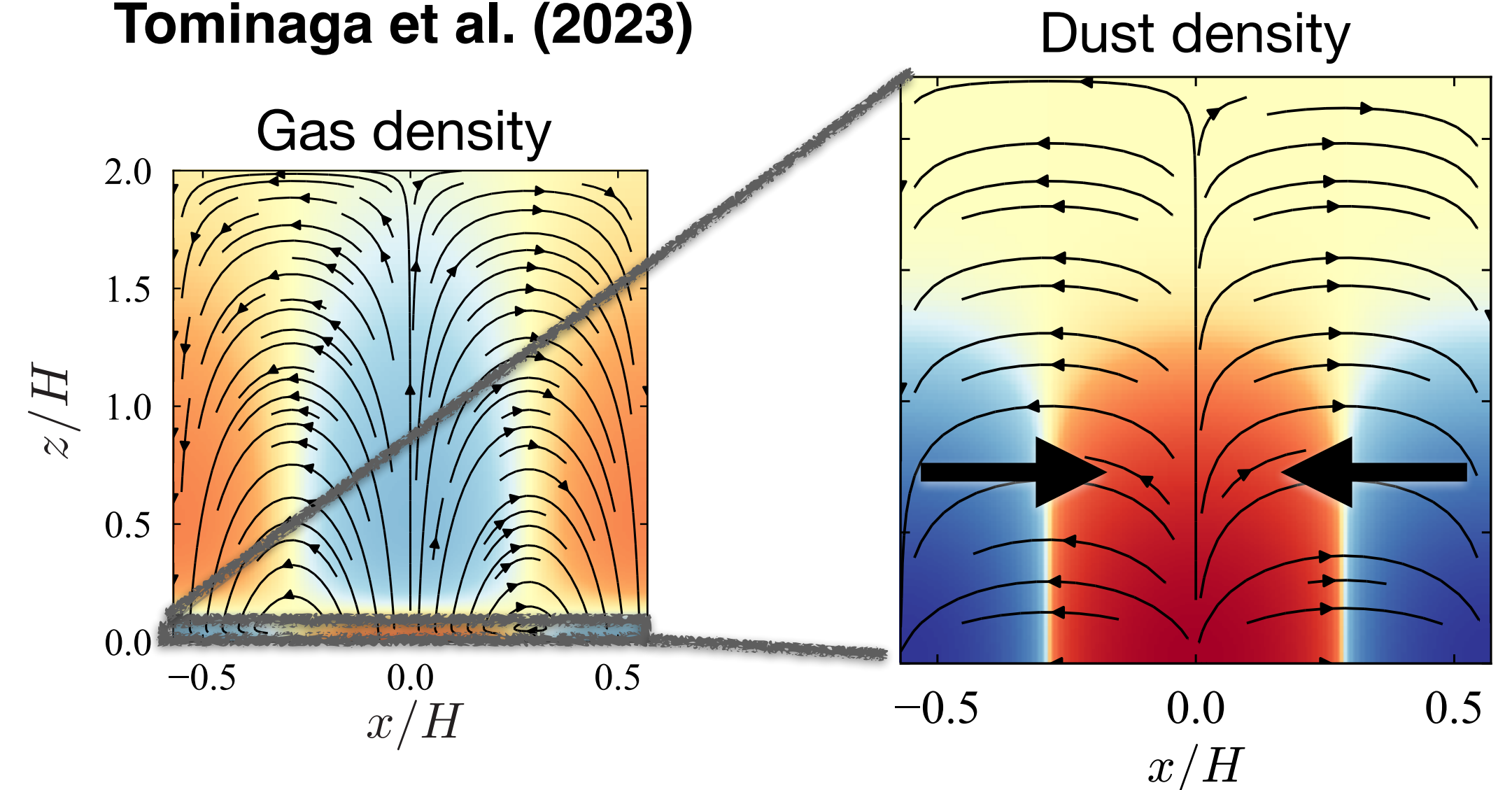
- Length scale < Gas scale height
- Inducing turbulence/clumping

(e.g., Youdin & Goodman 2005; Johansen et al. 2007; Simon et al. 2016)

Secular gravitational instability

(mode distinct from classical GI)

Tominaga et al. (2023)

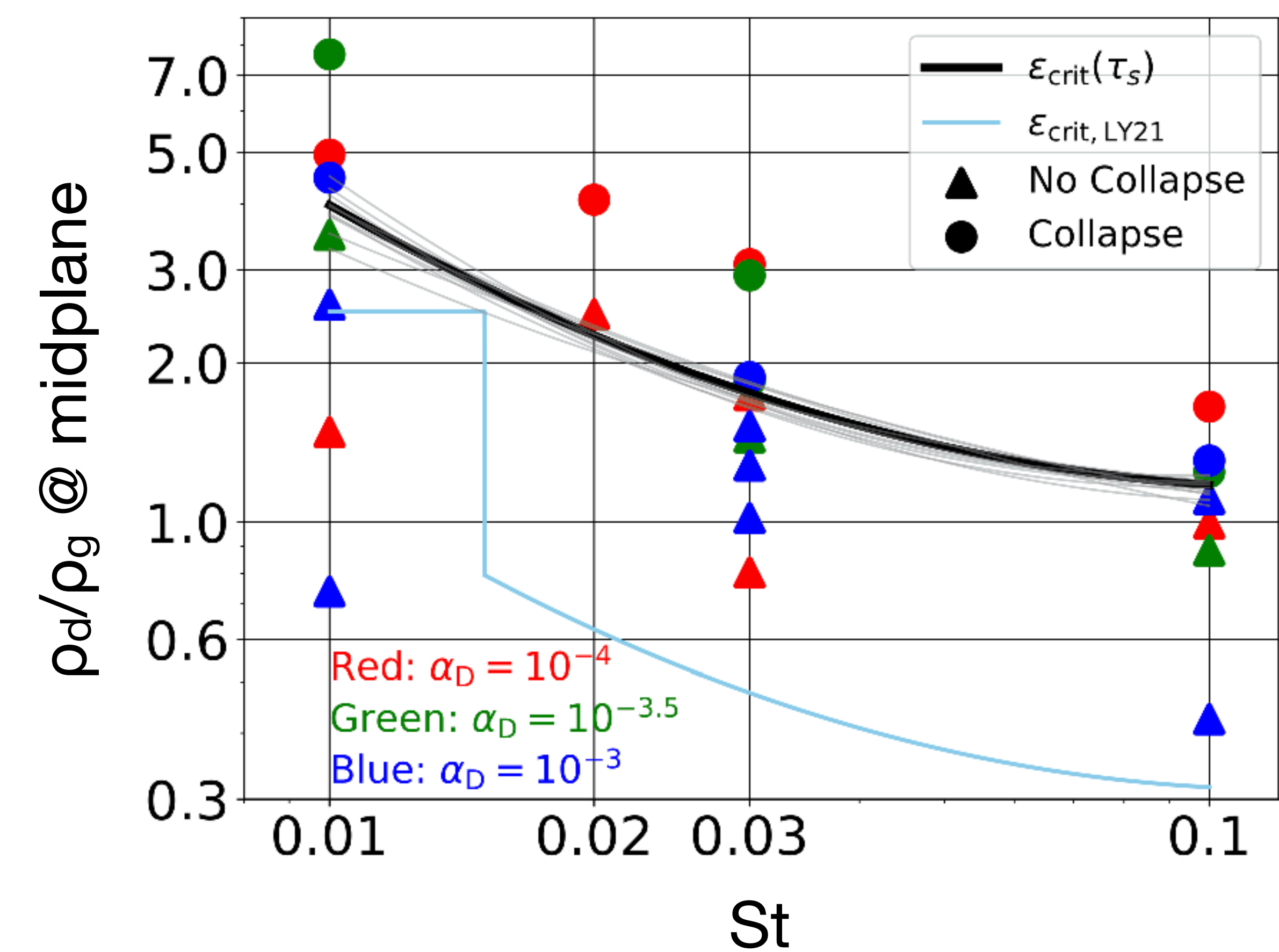


- Length scale $\sim$ Gas scale height
- Dust ring formation @ GI-**stable** disks ($Q > 1$)

see, Yamaguchi, Machida, **Tominaga et al. (2026)**
for **observations** of dust rings formed by secular GI

(e.g., Youdin 2011; **Tominaga et al. 2018,2019,2020,2023**)

Under what conditions does SI cause planetesimal formation?



(adapted from Lim et al. 2024)

$$St \equiv t_{stop} \Omega \sim 0.1 \left(\frac{a}{1 \text{ cm}} \right) \left(\frac{\Sigma_g}{10 \text{ g/cm}^2} \right)^{-1}$$

- Large dust grains & high ρ_d/ρ_g for efficient planetesimal formation
- Strong clumping seems to occur even for small St (e.g., ~ 0.01).

(e.g., Carrera et al. 2015; Yang et al. 2017; Li & Youdin 2021)

Clumping via SI?

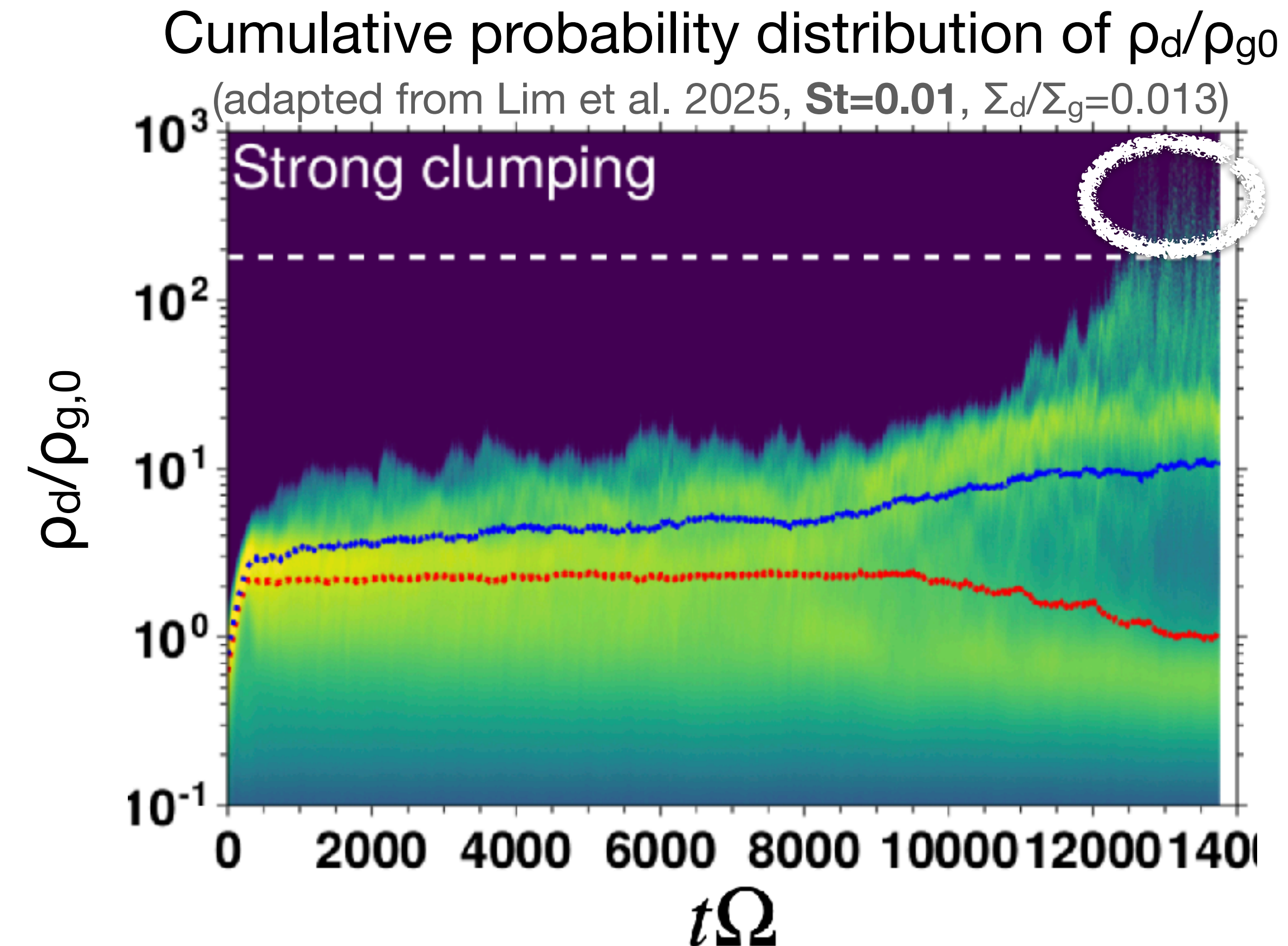
Drift barrier (e.g., $St \sim 0.1$)

Fragmentation barrier

Gap between the dust-growth and SI-based models

- If dust grains are relatively small (e.g., $St < 0.1$),
 - total clump mass is small,
(e.g., see Johansen & Youdin 2007, Bai & Stone 2010, and Lim et al. 2025 for density PDF/CDF)
 - clumping may be too slow.
(e.g., Carrera & Simon 2022 for SI around a pressure bump)
- Fragmentation may keep dust grains small.
(e.g., Birnstiel et al. 2009)

$$St_{\text{frag}} \sim 10^{-2} \left(\frac{v_{\text{frag}}}{1 \text{ m/s}} \right)^2 \left(\frac{T}{160 \text{ K}} \right)^{-1} \left(\frac{\alpha}{10^{-4}} \right)^{-1}$$



How do dust grains grow to the size of $St \sim 0.1-1$ before the strong clumping?

Dust growth in the pre-clumping phase may help?

- Dust grains may grow in the SI turbulence because of low-speed collisions @ dust-rich regions.

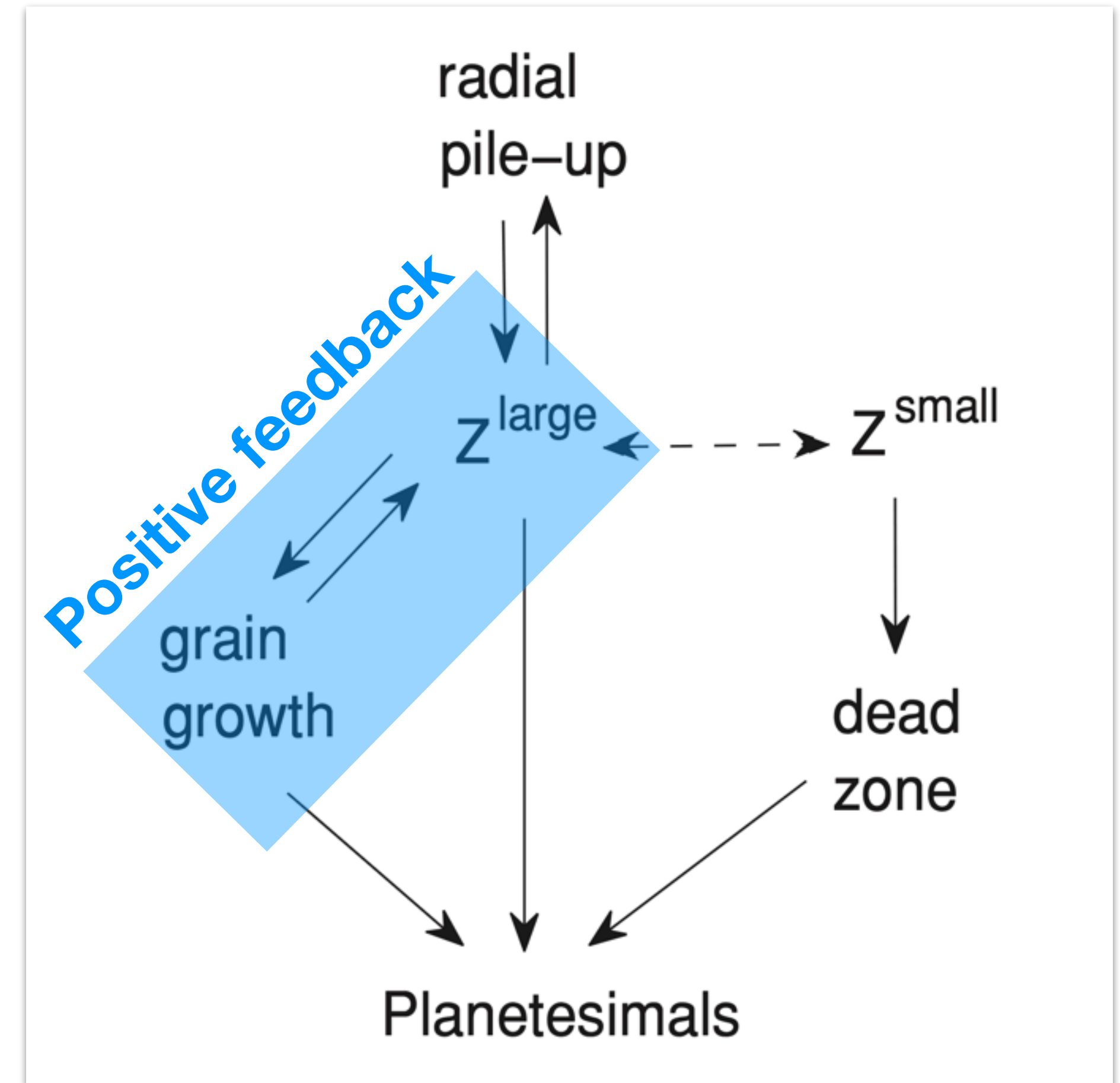
(Johansen et al. 2009; Bai & Stone 2010)

- Positive feedback loop of dust growth & clumping?

Bai & Stone 2010: increase in Σ_d due to dust growth $\rightarrow$ further clumping

Carrera et al. 2025 (talk yesterday): turbulence dampened in SI filaments

$\rightarrow$ growth w/ reduced collision speeds



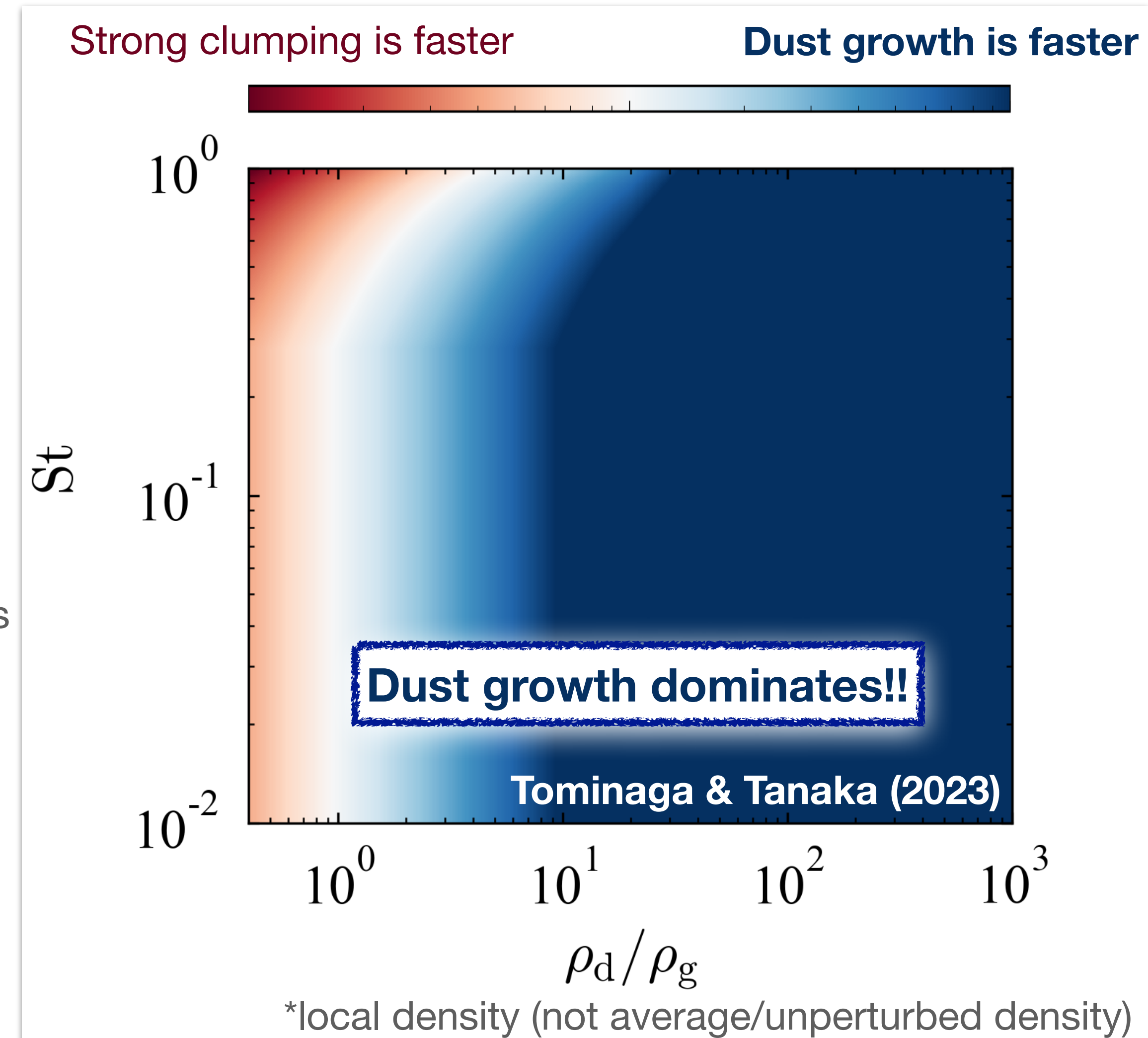
(adapted from Bai & Stone 2010)

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Carrera et al. 2025 (talk yesterday): turbulence dampened in SI filaments
 $\rightarrow$ growth w/ reduced collision speeds
- **Tominaga & Tanaka (2023):**
dust growth in the SI turbulence will be faster than the strong clumping!

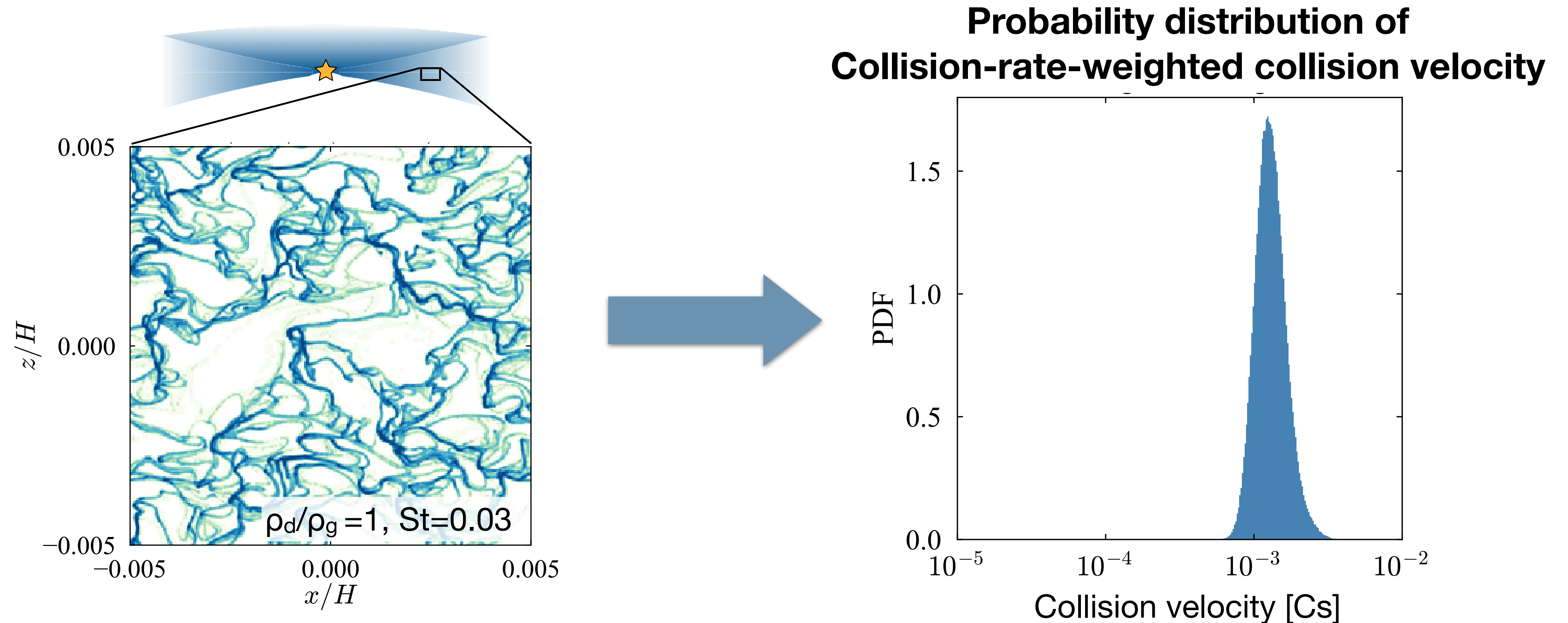
This study (Tominaga & Tanaka 2025)

We aim to reveal how **the SI-assisted coagulation** relaxes conditions for planetesimal formation by quantifying the collision velocity & collision rate in detail.

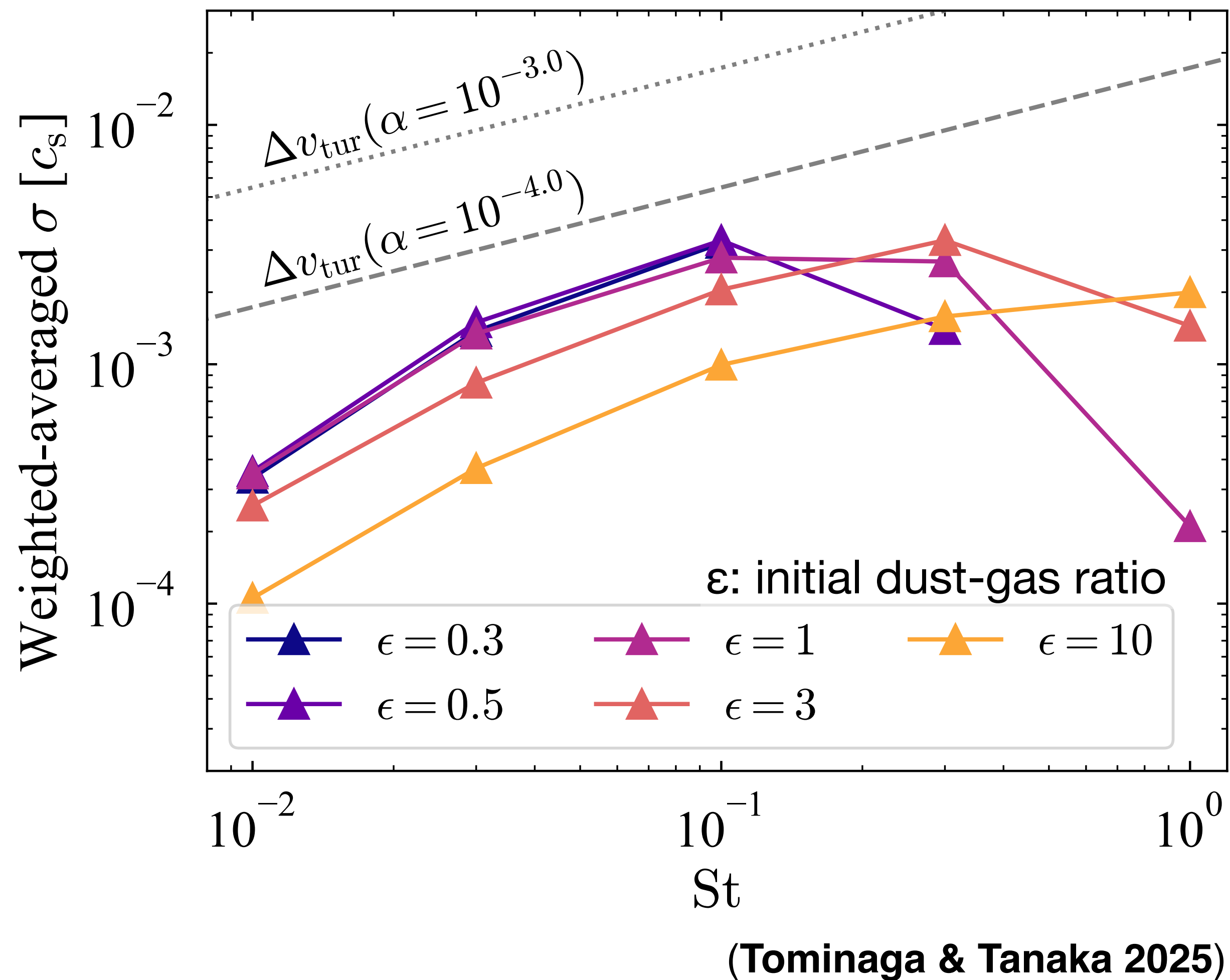


Tominaga & Tanaka (2025): Statistics of dust collisions in SI turbulence

- Local simulations with Athena w/ super-particle module (Stone et al. 2008; Bai & Stone 2010)
- We measure collision rates & velocities based on particle trajectories and ensemble average.



Typical collision velocity of dust grains



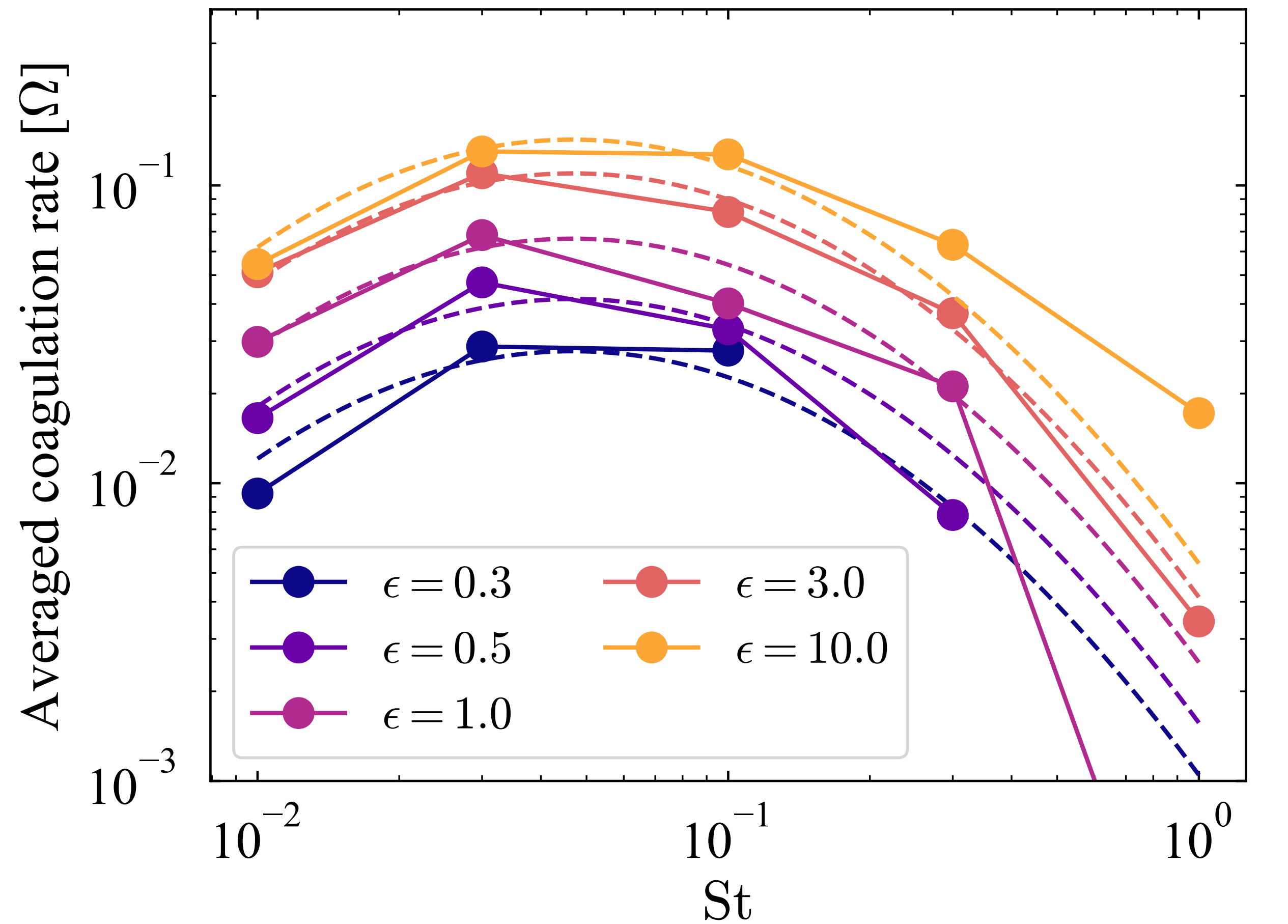
- Collision velocity increases as St increases but is still low ($\sim 10^{-3}c_s$).

$$10^{-3}c_s \sim 0.8 \text{ m/s} \left(\frac{T}{160 \text{ K}} \right)^{0.5}$$

→ In SI turbulence, dust grains can grow even if they are very fragile (e.g., $v_{\text{frag}} \sim 1 \text{ m/s}$)

- Mass loading decreases the collision velocities.
(see, Bai & Stone 2010, Schreiber & Klahr 2018; Carrera et al. 2025)

Averaged coagulation timescale (size-doubling timescale)

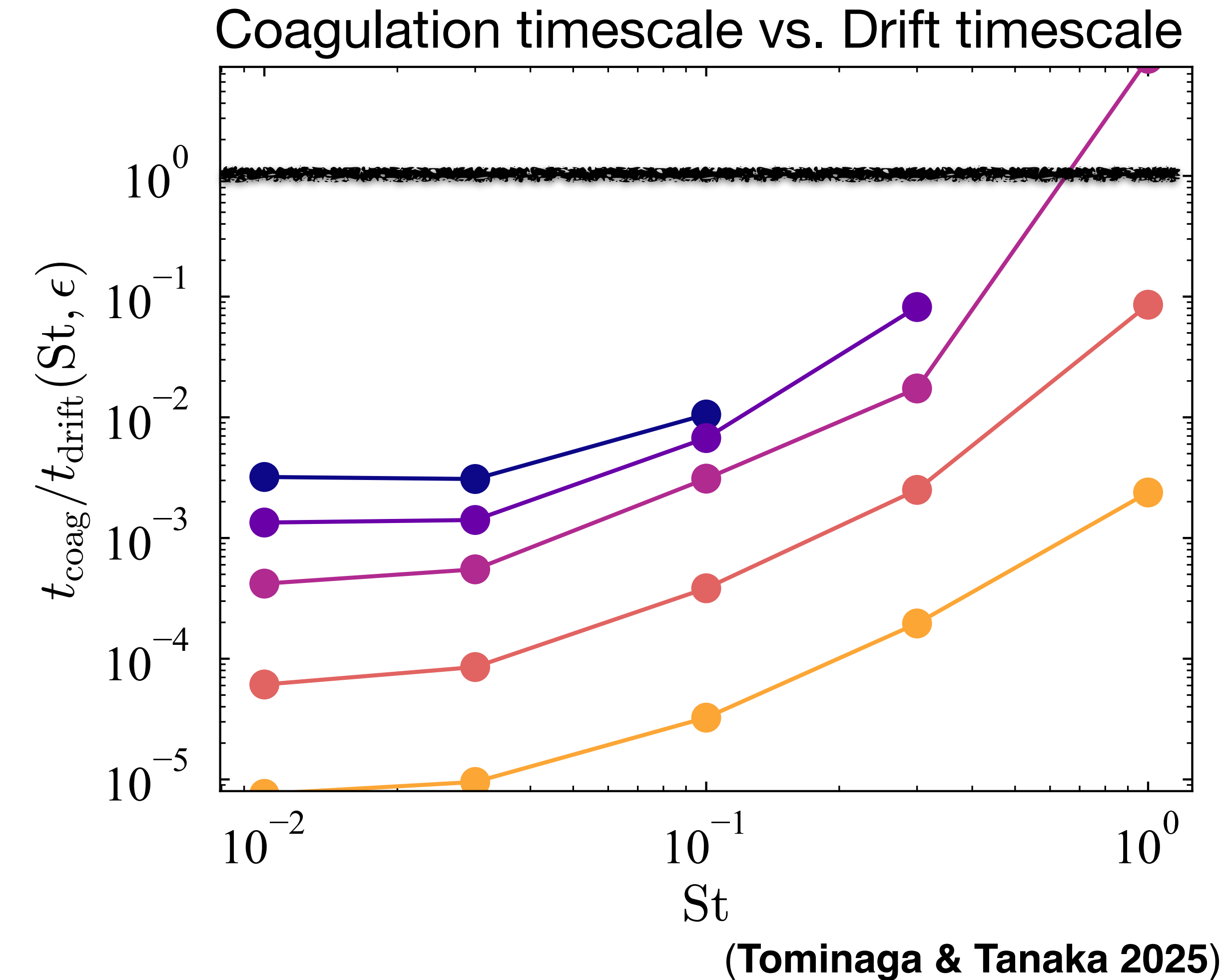


(Tominaga & Tanaka 2025)

dashed lines: empirical formulas

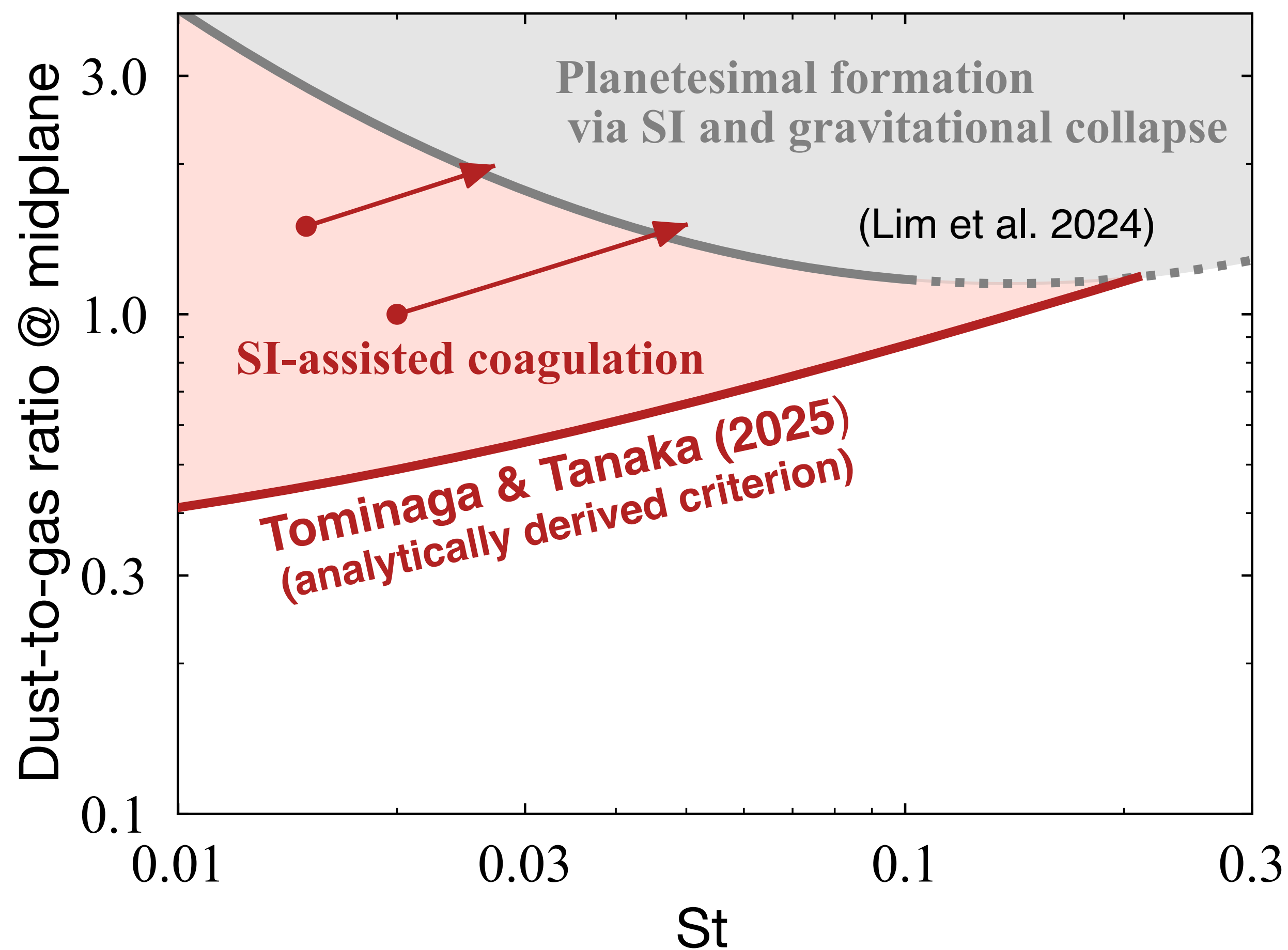
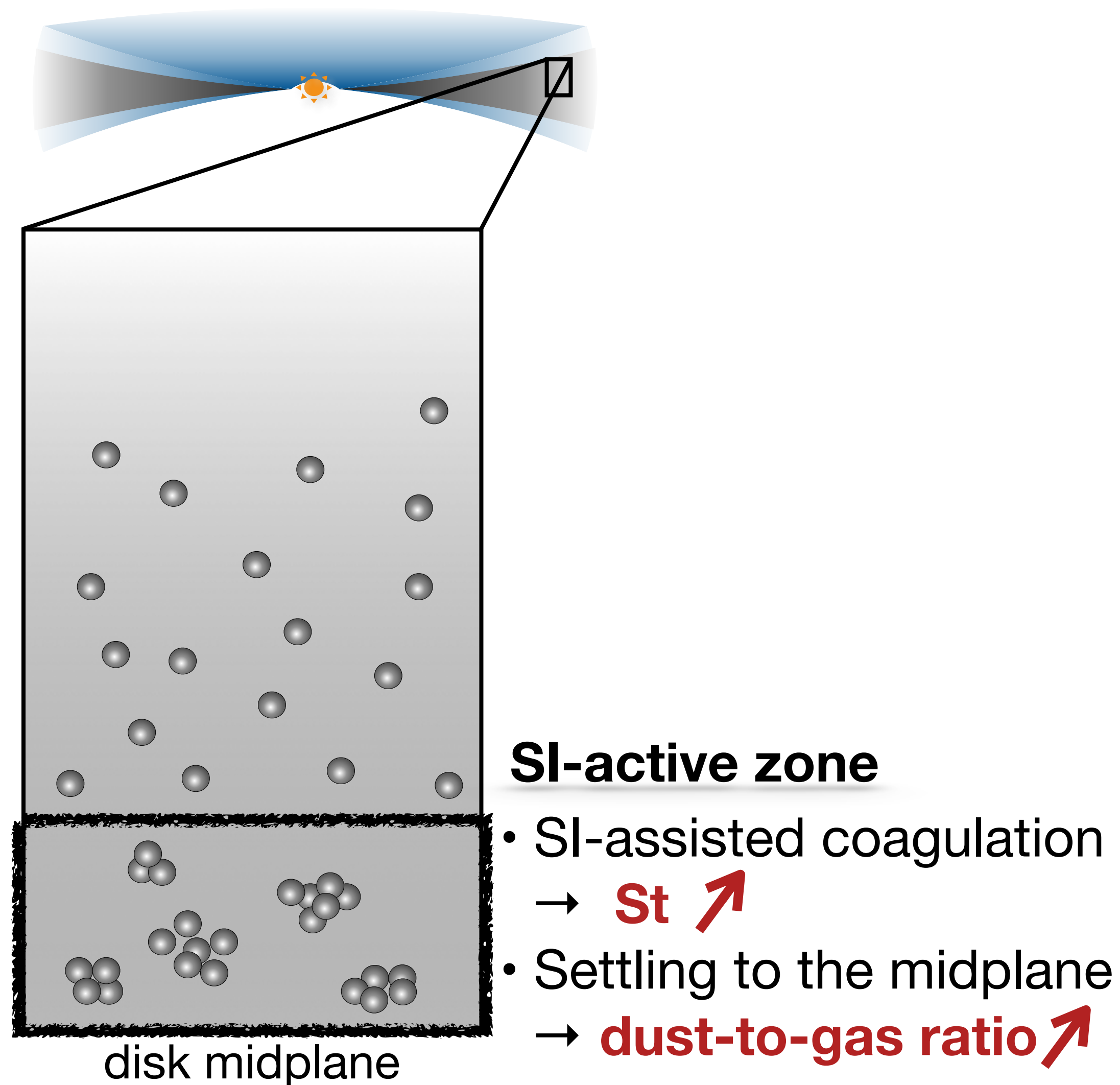
- Coagulation timescale is only ~ 2 - 20 T_{Kep} .
*The moderate clumping promotes collisions.

Averaged coagulation timescale (size-doubling timescale)



- Coagulation timescale is only $\sim 2\text{-}20 T_{\text{Kep}}$.
*The moderate clumping promotes collisions.
- **SI-assisted growth is much faster than the drift!**

SI-assisted growth promotes planetesimal formation!

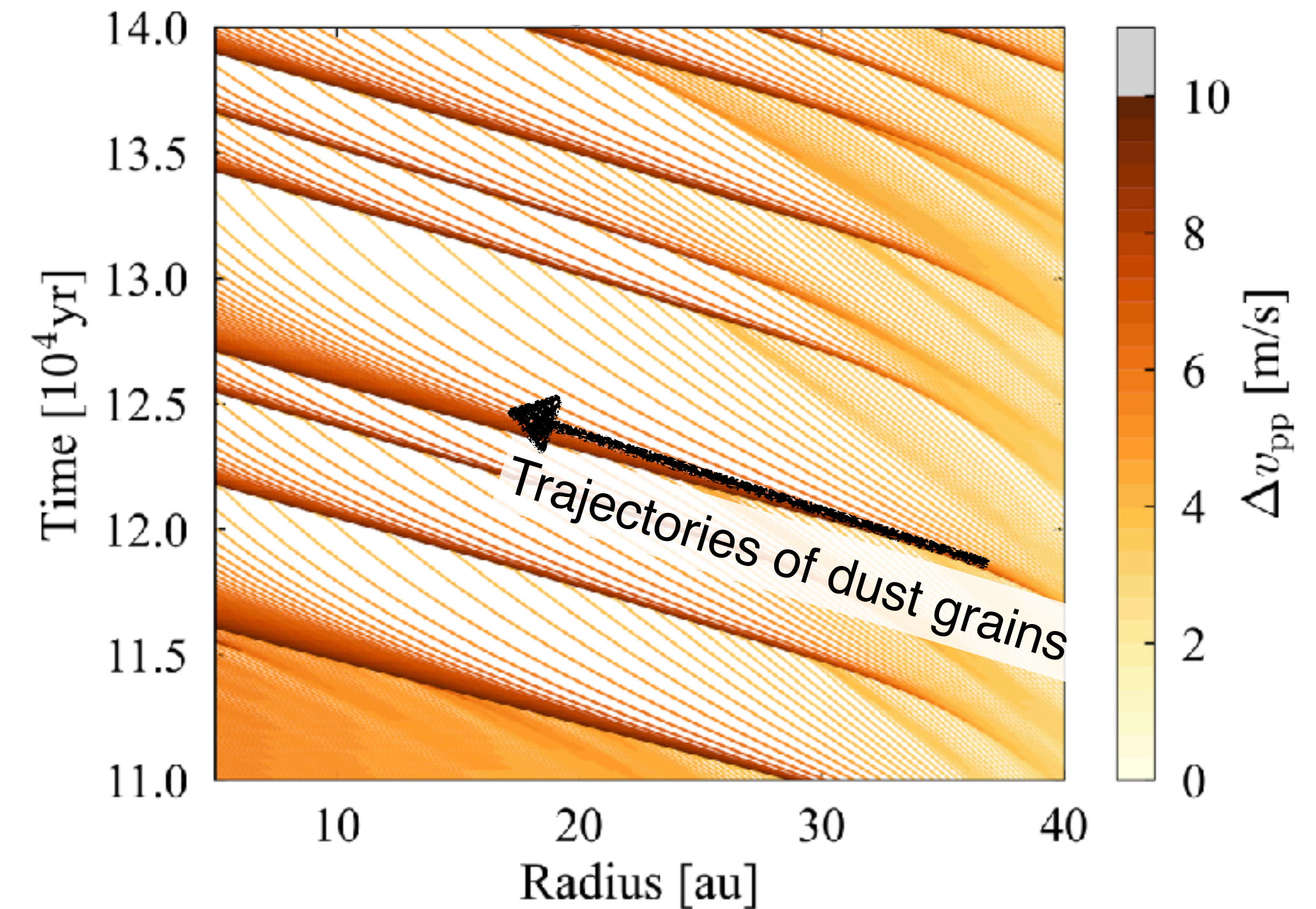
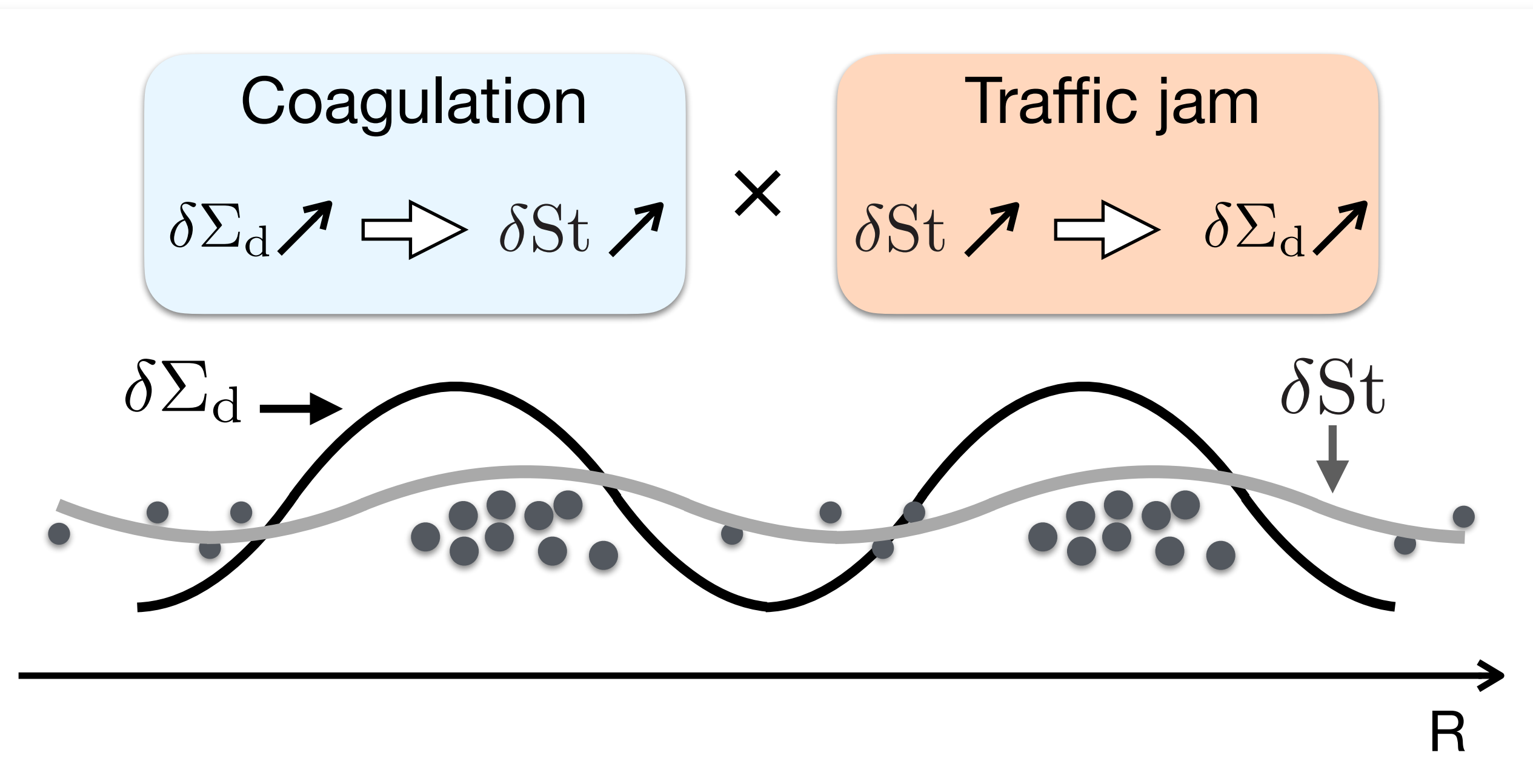


SI-assisted coagulation significantly relaxes conditions for planetesimal formation!

Implications: coagulation-driven dust concentration

Coagulation Instability (CI)

(Tominaga et al. 2021, 2022a,b)



- CI can develop if fragmentation does not impede dust growth. (e.g., $v_{\text{frag}} \geq 3$ m/s; see Tominaga et al. 2022b)
- If the SI-assisted coagulation relaxes the fragmentation barrier,
CI will develop and cause runaway dust concentration on larger scales ($\sim H$)! (see also, Pierens et al. 2026)
→ also triggering secular GI? (Future work)

Summary

- Planetesimal formation via SI requires very large dust grains ($St \geq 0.1$).
- We showed that SI-turbulence promotes dust growth at low collision speeds, which fills the gap between initial dust growth and planetesimal formation!
- The condition for planetesimal formation is appreciably relaxed by the SI-assisted coagulation.

